IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

STABILITY AND CONTROL OF NON-LINEAR SYSTEMS

Thursday 18 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : D. Angeli
Second Marker(s) : E.C. Kerrigan

© Imperial College London

1. Consider the following nonlinear bidimensional time-invariant autonomous system:

$$\begin{aligned} \dot{x}_1 &= -x_1 + \sin(2x_1) + x_2 \\ \dot{x}_2 &= -x_2 + \sin(2x_2) - x_1 \end{aligned}$$

with state $x = [x_1, x_2]'$ evolving in $\mathbb{R}^2$.

a) Discuss existence and uniqueness of solutions. [2]

b) Find analytically and sketch graphically the nullclines of the vector-field. [2]

c) Find regions in $\mathbb{R}^2$ delimited by nullclines and identify the orientation of the vector-field in each one of them. [2]

d) Compute all equilibria of the system, linearize the system around each equilibrium and discuss the local phase-portrait. [3]

e) Consider the following M-parameterized family of discs:

$$D_M := \{x : x_1^2 + x_2^2 - M \leq 0\}.$$

Show that for M sufficiently large the discs D_M are forward invariant for the dynamics. (Hint: use the fact that $|\sin(x)| \leq 1$ to bound the non-definite terms in the inequality needed to establish invariance). [4]

f) Use the results in e) to prove existence of a compact forward invariant region which does not contain equilibria. [3]

g) Discuss the ω-limit set $\omega(x_0)$ for $x_0 \neq 0$. [2]

h) Sketch a global phase-portrait for the system. [2]

2. Consider the following second-order differential equation:

$$\ddot{z}(t) = -(z^2(t) + \dot{z}^2(t))\dot{z}(t) - z^5(t) + u(t),$$

where $z(t)$ and $u(t) \in \mathbb{R}$.

a) Find a state-space representation of the system, with input $u(t)$ and output $y(t) := \dot{z}(t)$. [3]

b) Highlight that the system considered is input-affine. [2]

c) Show, by choosing a suitable storage function, that the system is passive from input u to output y. [4]

d) Consider the system dynamics for $u = 0$. Propose a candidate Lyapunov function $V(x)$ for assessing stability of the origin. (Hint: use the storage function as a candidate Lyapunov function). Show that $V(x)$ is positive definite and radially unbounded. [3]

e) Compute the derivative $\dot{V}(x)$, along solutions of the system (for $u = 0$), and discuss its sign definiteness. [2]

f) Show, through a suitable stability criterion, that 0 is a globally asymptotically stable equilibrium for the unforced system ($u = 0$). [3]

g) Consider next arbitrary constant values of u; find the corresponding equilibrium, and denote it as $x_e(u)$. How could you study the stability properties of $x_e(u)$ for the corresponding system, forced by a constant input u ? [3]

3. Consider the nonlinear bidimensional system defined below in state-space representation:

$$\begin{aligned} \dot{x}_1 &= \text{atan}(-2x_1 + x_2), \\ \dot{x}_2 &= (-x_2 + x_1)^3. \end{aligned}$$

a) Recast the system as a feedback interconnection of two nonlinear scalar systems Σ_1 and Σ_2. Write the "open-loop" equations of each system with corresponding input signals. [4]

b) Show that system Σ_1 is Input-to-State Stable (ISS) with respect to its scalar input, and compute the corresponding ISS gain (Hint: try to be as tight as possible). [4]

c) Show that system Σ_2 is Input-to-State Stable with respect to its scalar input, and compute the corresponding ISS gain. [4]

d) Exploit the previous analysis to argue that the system is globally asymptotically stable (GAS) at the origin. [4]

e) Would the conclusion in question d) still be true replacing $+x_1$ with $-x_1$ in the equation for $\dot{x}_2$? Why ? [4]

4. The following equations describe a nonlinear control system, with input $u \in \mathbb{R}$ and output $y \in \mathbb{R}$ and state $x \in \mathbb{R}^3$:

$$\begin{aligned} \dot{x}_1 &= \sin(x_2 - x_1) + \sin(x_3 - x_1) + u \\ \dot{x}_2 &= \sin(x_1 - x_2) + u \\ \dot{x}_3 &= \sin(x_1 - x_3) - x_3 \end{aligned}$$

$$y = x_1 - x_2$$

a) Compute the relative degree of the system at $x = 0$ and explain if it is locally or globally defined. [3]

b) Design an input-output linearizing state feedback law: $u(t) = k(x(t)) + v(t)$, where $v(t)$ is the new auxiliary input signal. [3]

c) Bring the system in normal form, and highlight the internal dynamics. (Hint: Choose a linear combination of state variables, linearly independent from the output). [6]

d) Discuss the Input-to-State Stability of the internal dynamics, when y is regarded as their input signal. [4]

e) Design a stabilizing feedback for the system around the equilibrium at the origin, and justify the answer provided. [2]

f) Would the conclusion of the analysis carried out so far be different if $-x_3$ in the equation for $\dot{x}_3$ is replaced by $+x_3$? Explain your answer. [2]